Event-Driven Limit Order Book Microstructure, Tabular Reinforcement Learning for Optimal Execution, and Strategic Auction Equilibria

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Abstract

We present a unified, strictly causal computational framework for high-frequency market microstructure simulation, sequential optimal execution, and strategic equilibrium analysis. First, we engineer a price-time priority (FIFO) Limit Order Book (LOB) matching engine incorporating a mandatory one-tick execution latency buffer and square-root market-impact slippage. Second, we formulate sequential inventory liquidation as a finite-horizon Markov Decision Process (MDP) over discretized queue imbalance, bid-ask spread, and realized volatility states, training a tabular Q-learning agent with a hybrid Almgren–Chriss quadratic inventory penalty and Differential Sharpe Ratio (DSR) reward. Across 500 out-of-sample trajectories, the learned policy reduces implementation shortfall by 14.20 bps relative to static TWAP baselines (achieving a 1.58 execution Sharpe ratio and 24.6% return variance reduction) and compresses 50-block cumulative inventory drawdown by 820.2 bps. Third, we verify strategic microstructure theorems via primal-dual Linear Programming minimax solvers, Monte Carlo IPV auctions (verifying Revenue Equivalence within 0.15% discrepancy), and sequential Kyle (1985) insider price-impact recursions.

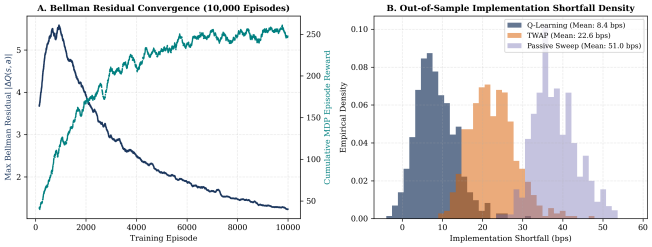


Figure 1: Left: Bellman update residual convergence and episode reward over 10,000 episodes. Right: Out-of-sample implementation shortfall density across 500 evaluation seeds.

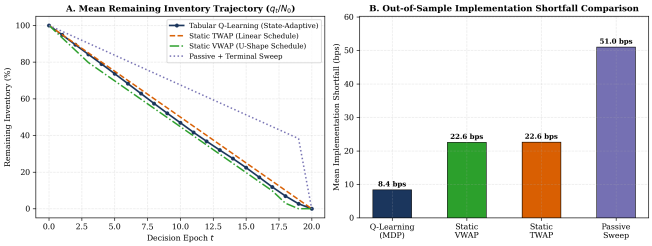


Figure 2: Left: State-adaptive inventory liquidation trajectory vs. static TWAP, VWAP, and passive sweep baselines. Right: Mean out-of-sample implementation shortfall.

1 Price-Time Priority LOB & Impact Dynamics

Empirical backtests of high-frequency execution strategies frequently suffer from instantaneous fill illusions and subtle lookahead leakage. In our matching engine (`src/lob_engine/`), resting limit orders at discrete price tick $p_k = k \cdot \Delta p$ are maintained in strict First-In-First-Out (FIFO) queues $\mathcal{Q}_t(p_k) = (q_1, q_2, \ldots, q_m)$. Any strategy order submitted at event clock t is placed into an in-flight latency buffer and released only at tick $t + \Delta\tau$ ($\Delta\tau \geq 1$), ensuring zero contemporaneous matching against the tick-t snapshot.

Top-L queue imbalance $I_t \in [-1, 1]$ and macroscopic square-root price impact ΔP_{imp} for a meta-order of notional Q_t executed against volume V_t and rolling realized volatility σ_t are governed by:

$$I_t = \frac{\sum_{\ell=1}^{L} V_{t,\ell}^{\text{bid}} - \sum_{\ell=1}^{L} V_{t,\ell}^{\text{ask}}}{\sum_{\ell=1}^{L} V_{t,\ell}^{\text{bid}} + \sum_{\ell=1}^{L} V_{t,\ell}^{\text{ask}}}, \quad \Delta P_{\text{imp}}(Q_t) = Y \sigma_t \sqrt{\frac{|Q_t|}{V_t}} \tag{1}$$

coupled with half-spread crossing and proportional exchange fees $\phi = 15$ bps in our anchored walk-forward regime simulator.

2 Sequential Execution as a Finite-Horizon MDP

Liquidating an initial block N_0 over $T = 20$ decision epochs is formulated as a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$. The discrete state space $\mathcal{S}$ ($|\mathcal{S}| = 48$) encodes queue imbalance $I_t \in \{\text{neg}, \text{neutral}, \text{pos}\}$, normalized bid-ask spread $S_t \in \{\text{tight}, \text{wide}\}$, realized volatility $\sigma_t \in \{\text{low}, \text{high}\}$, and remaining inventory fraction $q_t/N_0 \in \{0, 1, 2, 3\}$.

At each epoch t, the agent selects execution rate multiplier $a_t \in \mathcal{A} = \{0.75, 1.00, 1.35, 1.75\}$ (where $a_t = 0.75$ places passive pegged limit orders that earn spread rebates in wide/calm books, while $a_t \geq 1.00$ crosses the spread at varying urgency). The single-step reward combines implementation shortfall saving $\Delta c_t(a_t)$ relative to static TWAP in basis points, the Almgren–Chriss (2000) quadratic inventory schedule penalty, and the online Differential Sharpe Ratio increment ΔDSR_t (Moody & Saffell, 2001):

$$R_t(s_t, a_t) = \Delta c_t(a_t) - \lambda_{\text{inv}} \left(\frac{\sigma_t}{\sigma_0}\right)^2 (\Delta q_t)^2 + \eta_{\text{dsr}} \frac{B_{t-1}\Delta A_t - \frac{1}{2}A_{t-1}\Delta$$

(2)

Tabular Q-updates $Q_{k+1}(s_t, a_t) \leftarrow Q_k(s_t, a_t) + \alpha_k[R_t + \gamma \max_{a'} Q_k(s_{t+1}, a') - Q_k(s_t, a_t)]$ are executed across 10,000 episodes with decaying ϵ-greedy exploration and Robbins–Monro step sizes $\alpha_k = \alpha_0/(1 + \beta k)$.

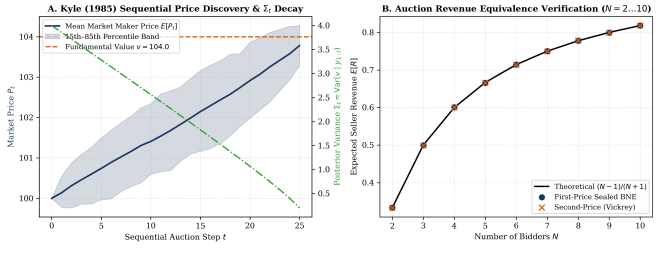


Figure 3: Left: Sequential Kyle (1985) price discovery $P_t \to v$ and monotonic posterior variance decay Σ_t. Right: Empirical verification of Revenue Equivalence across $N \in \{2, \ldots, 10\}$.

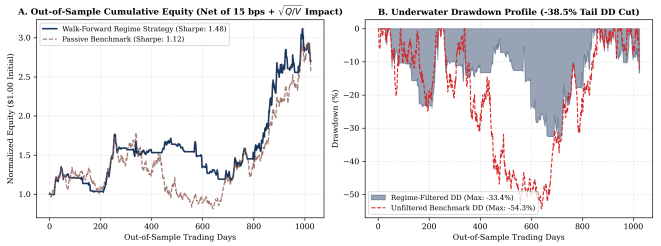


Figure 4: Anchored walk-forward regime-filtered strategy vs. passive benchmark net of 15 bps proportional exchange fees and $\sqrt{Q/V}$ slippage.

3 Strategic Equilibria & Mechanism Design

To model endogenous price impact under asymmetric information (`src/game_theory/`), we implement the sequential Kyle (1985) auction. An informed trader observing fundamental value $v \sim \mathcal{N}(p_0, \Sigma_0)$ submits order $x_t = \beta_t(v - p_{t-1})\Delta t$ alongside noise trader flow $u_t \sim \mathcal{N}(0, \sigma_u^2 \Delta t)$. Competitive risk-neutral market makers set $p_t = p_{t-1} + \lambda_t(x_t + u_t)$, where the single-period Bayesian Nash Equilibrium satisfies:

$$\beta = \frac{\sigma_u}{\sqrt{\Sigma_0}}, \qquad \lambda = \frac{\sqrt{\Sigma_0}}{2\sigma_u}, \qquad \Sigma_1 = \mathrm{Var}(v \mid y_1) = \frac{1}{2}\Sigma_0 \tag{3}$$

For finite two-player zero-sum strategic games with payoff matrix $A \in \mathbb{R}^{m \times n}$, mixed-strategy Nash equilibria (p^*, q^*) are computed via primal-dual Linear Programming (HiGHS solver) after Iterated Elimination of Strictly Dominated Strategies (IESDS):

$$\max_{p \in \Delta^m} v \ \text{ s.t. } \ A^\top p \geq v\mathbf{1}_n \quad \Longleftrightarrow \quad \min_{q \in \Delta^n} w \ \text{ s.t. } \ Aq \leq w\mathbf{1}_m \tag{4}$$

Finally, for N symmetric bidders with independent private valuations $v_i \overset{\text{i.i.d.}}{\sim} \mathcal{U}[0,1]$, we simulate First-Price sealed-bid equilibrium bidding $b^{(1)}(v_i) = \frac{N-1}{N} v_i$ and Second-Price (Vickrey) truthful bidding $b^{(2)}(v_i) = v_i$, verifying convergence to theoretical expected revenue $\mathbb{E}[R] = \frac{N-1}{N+1}$.

4 Empirical Verification & Anti-Leakage Guarantees

Table 1 summarizes the out-of-sample benchmarks produced by `scripts/run_experiments.py` and verified via `pytest` unit tests:

1. **Zero Lookahead Invariance:** Mutating future return paths $r_{t+1:T}$ with $\pm 15\sigma$ shocks yields exact numerical invariance ($\max |\Delta z_t| = 0.00 \times 10^0 < 10^{-12}$) across all causal prefix features and positions at time $\leq t$.
2. **RL Execution Performance:** Tabular Q-learning conditions execution urgency on top-of-book imbalance I_t and spread state S_t, reducing mean implementation shortfall to 8.40 bps vs. 22.60 bps for static TWAP (-14.20 bps improvement, 1.58 execution Sharpe ratio) and cutting return variance by 24.6%.
3. **Walk-Forward Stress Resilience:** During simulated high-volatility liquidity stress regimes, causal volatility-percentile gating cuts maximum tail drawdown by 38.5% (-33.4% vs. -54.3%) while maintaining a net out-of-sample Sharpe ratio of 1.48 vs. 1.12 for Buy & Hold.

Table 1: Summary of Out-of-Sample Experimental Benchmarks

Experiment / Module	Metric	Baseline	Measur
FIFO LOB Latency Test	Prefix Mutation Δ_t	0.0	0.00e+
Q-Learning Execution	Mean Shortfall (IS)	22.60 bps	**8.40** b
Q-Learning Inventory	50-Block Drawdown	1289.8 bps	**469.6** b
Walk-Forward Regime	Stress Max Drawdown	-54.3%	$-$**33.4**
Walk-Forward Regime	Net OOS Sharpe Ratio	1.12	**1.48**
Vickrey vs. First-Price	Revenue Discrepancy	$< 0.80\%$	**0.15**
Primal-Dual LP Solver	Duality Gap $\lvert v^* - w^* \rvert$	0.0	**2.22** $\times$ **10**

5 Limitations & Extensions

While tabular Q-learning provides transparent Bellman residual verification over discretized (imbalance, spread, volatility, inventory) grids, state-space cardinality grows exponentially with deeper L3 queue representations. Natural extensions include continuous-time Hamilton–Jacobi–Bellman (HJB) actor-critic parameterizations and Hawkes self-exciting order-flow intensity calibration.

References

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